

Linear algebra – 1

Determinants and invertibility

Homework

12 сентября 2026 года

Theoretical minimum. The adjugate satisfies $A \operatorname{adj} A = (\det A)E$ and has rank n , 1 or 0 according to the rank of A ; Jacobi's formula differentiates a determinant and gives $\det e^A = e^{\operatorname{tr} A}$. A determinant of a matrix which is a small-rank perturbation of a simple one is computed by $\det(A + UV^T) = \det A \cdot \det(E + V^T A^{-1}U)$. An integer matrix is invertible over $\mathbb{Z}$ exactly when $\det A = \pm 1$, and since $\det$ is a polynomial with integer coefficients, identities transfer to any commutative ring and reduction modulo a prime proves non-vanishing. Singularity has three equivalent faces — dependent columns, a zero eigenvalue, a non-trivial kernel — and the art is to pick the right one. Finally, over $\mathbb{R}$ a determinant is a continuous function of the entries, so a determinant identity proved on a dense set holds everywhere: this is how one passes from invertible matrices to all of them.

1. (VJIMC 2013) Let $A = (a_{ij})$ and $B = (b_{ij})$ be two real 10×10 matrices such that $a_{ij} = b_{ij} + 1$ for all i, j and $A^3 = O$. Prove that $\det B = 0$.
2. (IMC 2015) For any integer $n \geq 2$ and two $n \times n$ matrices with real entries A, B satisfying $A^{-1} + B^{-1} = (A + B)^{-1}$, prove that $\det A = \det B$. Does the same conclusion hold for matrices with complex entries?
3. (IMC 2014) Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\operatorname{tr} M = a$ and $\det M = b$.
4. (Romania national olympiad 2017) Let $n \geq 2$ and k be natural numbers and let $A_1, \dots, A_k$ be symmetric real $n \times n$ matrices. Prove that the following are equivalent:

- 1) $\det(\sum_{i=1}^k A_i^2) = 0$;
- 2) $\det(\sum_{i=1}^k A_i B_i) = 0$ for all $B_1, \dots, B_k \in M_n(\mathbb{R})$.
5. (Romania national olympiad 1995) Let n, k be positive integers and let $A_i \in M_n(\mathbb{R})$, $i = 1, \dots, k$. Prove that $\det(\sum_{i=1}^k A_i^T A_i) \geq 0$.
6. (Romania national olympiad 2011) Let A be a singular real matrix of order $n \geq 2$ and let $\text{adj } A$ be its adjugate. Prove that $\text{tr}(\text{adj } A) \neq -1$ if and only if the matrix $E + \text{adj } A$ is invertible.
7. (Romania national olympiad 2008) Let A be an $n \times n$ matrix with complex entries. Prove that $A^{-1} = \overline{A}$ if and only if there exists an invertible complex matrix B with $A = B^{-1} \overline{B}$.
8. (Romania national olympiad 2025) Let n be a positive integer and let a, b be complex numbers with $a \neq 1$ and $b^k \neq 1$ for every $k \in \{1, 2, \dots, n\}$. The matrices $A, B \in M_n(\mathbb{C})$ satisfy $BA = aE + bAB$. Prove that A and B are invertible.
9. (Romania national olympiad 2012) Let $n \geq 2$ and $1 \leq k \leq n-1$. Prove that if $A \in M_n(\mathbb{C})$ has exactly k minors of order $n-1$ equal to zero, then $\det A \neq 0$.
10. (Romania national olympiad 2014) Let $A \in M_4(\mathbb{R})$ be invertible with $\text{tr } A = \text{tr}(\text{adj } A) \neq 0$. Show that $A^2 + E$ is singular if and only if there is a non-zero matrix in $M_4(\mathbb{R})$ anticommuting with A .
11. (VJIMC 1991) Compute the determinant of an antisymmetric matrix $(a_{ij})_{i,j=1}^n$, where $a_{ij} = -a_{ji}$ and $a_{ii} = 0$, for odd n , and explain what changes for even n .
12. (Mexmat, 2007) Prove Carleman's inequality: for every real matrix $A = (a_{ij})_{i,j \leq n}$

$$|\det(E + A)| \leq \exp\left(\text{tr } A + \frac{1}{2} \sum_{i,j=1}^n a_{ij}^2\right).$$
13. Prove that if all entries of an integer $n \times n$ matrix are odd, then 2^{n-1} divides its determinant, and show that the exponent $n-1$ cannot be improved.

14. (Putnam 2008) Alan and Barbara play a game in which they take turns filling entries of an initially empty 2008×2008 array. Alan plays first. At each turn a player chooses a real number and places it in a vacant entry. The game ends when all the entries are filled. Alan wins if the determinant of the resulting matrix is non-zero, Barbara wins if it is zero. Which player has a winning strategy?
15. (Putnam 2002) In Determinant Tic-Tac-Toe, Player 1 enters a 1 in an empty 3×3 matrix, Player 0 counters with a 0 in a vacant position, and play continues in turn until the matrix is filled with five ones and four zeros. Player 0 wins if the determinant is 0 and Player 1 wins otherwise. Assuming optimal play, who wins and how?
16. (CIIM 2009) Determine whether for every natural n there is a real $n \times n$ matrix whose determinant is zero but such that changing any single entry produces a matrix with non-zero determinant.